This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Touch Display
Manufacturer: BenQ
Firmware Version: N/A
Model(s): RM8601K, RM7501K, RM6501K, RM5501K

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
3.03.0000-b006	2.3.0

Version History

Module Version	Date	Notes
1_0_0_0	6/26/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  `Example: InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  `Example: InterfaceName.connectionCounter = 5`

- Serial Only: DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '98', and 'Broadcast'. If Device ID is set to 'Broadcast', status is unavailable.
  `Example: InterfaceName.DeviceID = '1'`

- Etheret is hardcoded to use DeviceID of '1'

- For Ethernet connection, only Power Command will be sent to the device when it is Off.

- For Serial connection, only Power Command will be sent to the device when it is Android Off.

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='RM8601K')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='RM8601K')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4660, Model='RM8601K')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio [1]	'Default' 'Auto' '14:9'	'16:9' 'Panorama Mode' 'PC Mode'	'4:3' 'Just Scan'

```
# AspectRatio example
InterfaceName.Set('AspectRatio', 'Default')
```

Command	Value	Value
ButtonandIRControl [1]	'Enable'	'Disable'

```
# ButtonandIRControl example
InterfaceName.Set('ButtonandIRControl', 'Enable')
```

Command	Value	Value	Value
Input	'VGA' 'HDMI 3' 'OPS'	'HDMI 1' 'HDMI 4'	'HDMI 2' 'Android'

```
# Input example
InterfaceName.Set('Input', 'VGA')
```

Command	Value
Mute	'Toggle'

```
# Mute example
InterfaceName.Set('Mute', 'Toggle')
```

Command	Value	Value	Value
Power	'On'	'Off'	'Android Off'

```
# Power example
InterfaceName.Set('Power', 'On')
```

Command	Value	Value	Value
RemoteControl [2]	'Up' 'Right' 'Source' 'Freeze'	'Down' 'Ok' 'Back'	'Left' 'OSD Menu' 'Blank'

```
# RemoteControl example
InterfaceName.Set('RemoteControl', 'Up')
```

Command	Value	Value
Volume	'Up'	'Down'

```
# Volume example
InterfaceName.Set('Volume', 'Up')
```

[1] Only available for Ethernet Control.

[2] Only available for Serial Control.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
        InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
        Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
        InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
FeedbackHandler)
        FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
        InterfaceName.Update(Command)
        Value = InterfaceName.ReadStatus(Command)
        InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
        FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio[1]	'Default'	'16:9'	'4:3'
	'Auto'	'Panorama Mode'	'Just Scan'
	'14:9'	'PC Mode'	

```
InterfaceName.Update('AspectRatio')
Value = InterfaceName.ReadStatus('AspectRatio')
InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)
```

Command	Value	Value
ButtonandIRControl	'Enable'	'Disable'

```
InterfaceName.Update('ButtonandIRControl')
Value = InterfaceName.ReadStatus('ButtonandIRControl')
InterfaceName.SubscribeStatus('ButtonandIRControl', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'VGA'	'HDMI 1'	'HDMI 2'
	'HDMI 3'	'HDMI 4'	'Android'
	'OPS'		

```
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value	Value
Mute	'Toggle'	'On'	'Off'

```
InterfaceName.Update('Mute')
Value = InterfaceName.ReadStatus('Mute')
InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)
```

Command	Value
OperationHours	Hours

```
InterfaceName.Update('OperationHours')
Value = InterfaceName.ReadStatus('OperationHours')
InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)
```

Command	Value	Value	Value
Power	'On'	'Off'	'Android Off'

```
InterfaceName.Update('Power')
Value = InterfaceName.ReadStatus('Power')
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command **VolumeStatus**	Value 0 – 100

```
InterfaceName.Update('VolumeStatus')
Value = InterfaceName.ReadStatus('VolumeStatus')
InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)
```

[1] Only available for Ethernet Control.

[2] Only available for Serial Control.

Cable and Adapter Requirements

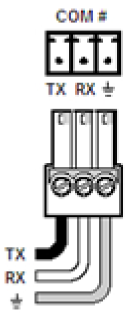
Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

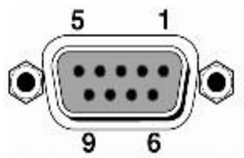
Serial communication

Port Type: RS-232
Baud Rate: 38400
Data Bits: 8
Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	⟶	2	RxD
RxD	⟵	3	TxD
GND	——	5	GND



RS232 DB9, Female Display

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type:	Ethernet
Default Port:	4660
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 14:9	801s1006\x0D
Aspect Ratio 16:9	801s1001\x0D
Aspect Ratio 4:3	801s1002\x0D
Aspect Ratio Auto	801s1003\x0D
Aspect Ratio Default	801s1000\x0D
Aspect Ratio Just Scan	801s1005\x0D
Aspect Ratio Panorama Mode	801s1004\x0D
Aspect Ratio PC Mode	801s1007\x0D
Button and IR Control Disable	801sC000\x0D
Button and IR Control Enable	801sC001\x0D
Input Android	801s"101\x0D
Input HDMI 1	801s"001\x0D
Input HDMI 2	801s"002\x0D
Input HDMI 3	801s"021\x0D
Input HDMI 4	801s"022\x0D
Input OPS	801s"102\x0D
Input VGA	801s"000\x0D
Mute Toggle	801s6002\x0D
Power Android Off	801s!002\x0D
Power Off	801s!000\x0D
Power On	801s!001\x0D
Remote Control Back	801s@023\x0D
Remote Control Blank	801s@031\x0D
Remote Control Down	801s@011\x0D
Remote Control Freeze	801s@032\x0D
Remote Control Left	801s@012\x0D
Remote Control Ok	801s@014\x0D
Remote Control OSD Menu	801s@020\x0D
Remote Control Right	801s@013\x0D
Remote Control Source	801s@021\x0D
Remote Control Up	801s@010\x0D
Volume Down	801s5200\x0D
Volume Up	801s5300\x0D

Appendix B. Update Commands

Aspect Ratio	`801gw000\x0D`
Button and IR Control	`801gi000\x0D`
Input	`801gj000\x0D`
Mute	`801gg000\x0D`
Operation Hours	`:01gv00000\x0D`
Power	`801gl000\x0D`
Volume Status	`801gf000\x0D`